

Energy Valve

Savings Estimator

How to drive the tool tab by tab: every dropdown, every field you can type in, and what every number it hands back actually means — with particular attention to the financial results, which are the ones most often misread.

Tool version	V.2026.11.1
Workbook	Belimo Energy Valve Estimator Tool.xlsm
Languages	English · Español · Português · Français
Units	US or Metric
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Belimo Americas · Internal field reference

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1 What the tool does

One sentence: it estimates what a building saves when Energy Valves let the chilled-water plant run at a higher Delta-T.

Every chilled-water plant is sized around a **design Delta-T** – the temperature rise between supply and return. Most plants never achieve it. They run at a lower Delta-T, which means that to deliver the same cooling they must push more water around the building.

That costs money twice over, and the tool separates the two:

SAVINGS STREAM	WHY IT HAPPENS
Pumping savings	Higher Delta-T means less flow for the same load. Pump power falls roughly with the cube of flow, so a modest flow reduction is a large power reduction.
Chiller savings	Warmer return water and better loading let the chiller plant run more efficiently – fewer machines on, better kW/ton.

The tool models the plant across four load bins over a full year, prices the energy at your electricity rate, and then puts a financial case around the difference. The tabs run left to right in the order you should fill them in.

The screenshots in this manual show a real worked example - a 1,500 ton, four-chiller plant lifting its Delta-T from 8 °F to 14 °F for a \$170,000 incremental investment. Your own numbers will differ; what matters is that every figure shown traces back to inputs you control.

2 Before you start

Language and units

Top-left of the Home tab. Both are dropdowns and both change the entire workbook the moment you pick them — labels, headers, unit suffixes and every converted number, on every tab.

CONTROL	CHOICES	NOTES
Language	English, Español, Português, Français	Translates labels, headings and the written summaries on the Results tab.
Units	US, Metric	Switches every flow, head, temperature, power and volume figure. The calculation is always done in US units and converted for display, so switching back and forth never drifts.
Save Settings	tick box	Ticked, the workbook reopens in the language and units you left it in. Unticked, it resets to English / US on every open.

Which cells can I type in?

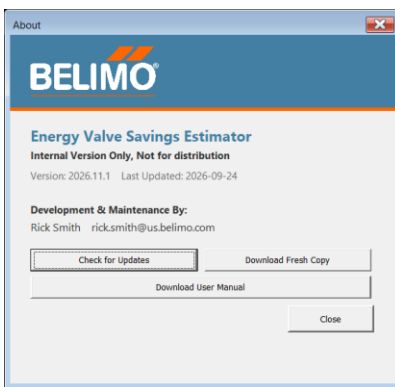
The workbook tells you, and the convention is the same on every tab. The Energy, Chiller and Pump tabs print the key on the sheet itself:

SHADING	MEANING
Pale / near-white	Editable field. Type here.
Blue-grey	Label or calculated result. Protected — the tool writes it.
Orange band	Section heading.

Sheets are protected so a stray keystroke cannot break a formula. If a cell will not accept typing, that is the tool telling you it is a result, not an input.

About screen and updates

Click the round blue (i) button at the top right of the Home tab, beside the red PDF Download button, to open the About screen. It shows the tool's name, its version and last-updated date, the internal-use notice, and who maintains the tool. It also has three buttons and a Close button.



The About screen, opened from the blue (i) button on the Home tab.

BUTTON	WHAT IT DOES
Check for Updates	Asks the update source whether a newer release exists. If one does, you are told what is new and may choose to download it.
Download Fresh Copy	Downloads the current published release straight away, even if you are already up to date – useful if you think your copy is damaged.
Download User Manual	Saves the current published copy of this manual (a PDF) to a folder you choose. If none has been published yet, the tool tells you so.
Close	Closes the About screen.

Each time the workbook opens it quietly checks Belimo's update source for a newer release. When one is available you are offered a summary of what changed. If you choose to install it, you pick where the new copy is saved through a normal Windows Save As dialog, just like any other file – your current copy is never changed or deleted. If the update source cannot be reached (for example you are offline), the check does nothing and the tool works as normal.

3 Home tab

The only tab most jobs need. Plant inputs at the top, the Delta-T comparison table in the middle, headline financials and greenhouse-gas equivalents below.

BELIMO

HomeEnergyChillerPumpFinancials

Belimo Energy Valve™
Savings Estimator



LanguageEnglishUnitsUS☐ Save Settings

ProjectUTSA

Information4 chillers; installed 1600 tons

Climate regionSouthChiller coolingWater CooledPlant designPrimary/Secondary

Utility CompanyNoneProgram

INPUT DATA: CHILLED WATER PLANT

	Value	Unit
Electricity cost (average)	0.12	\$/kWh
Design cooling load	1500	tons
Number of chillers	4	pcs.
Operating hours per day	24	hours
Operating days per week	7	days
Chiller design efficiency	0.63	kW/ton
Actual Secondary Pump Head (Design)	80	ft
Pump, VFD and motor efficiency	80	%
Design Delta-T	12	°F
Actual Delta-T before Energy Valve	8	°F
Future Energy Valve Delta-T setting	14	°F

ANNUAL ENERGY COSTS

Delta-T [°F]	Avg [GPM]	Avg [tons]	Pumping Cost	Chiller Cost	Total Cost	Energy [kWh]
6	2475	619	\$ 70,362	\$ 294,654	\$ 365,016	3,041,800
8	2006	669	\$ 41,411	\$ 249,763	\$ 291,174	2,426,450
10	1736	723	\$ 27,589	\$ 240,071	\$ 267,660	2,230,501
12	1534	767	\$ 18,763	\$ 195,381	\$ 214,144	1,784,536
14	1368	798	\$ 12,397	\$ 189,420	\$ 201,817	1,681,810
Annual Energy Savings for changing Delta T from 8 to 14			70% \$29,014	24% \$60,342	31% \$89,357	% (744,639)

INPUT FINANCIALS

	Value	Unit
Incremental Energy Valve investment	170,000	\$
Discount rate	4	%
Annual energy-cost inflation	2	%

FINANCIAL ANALYSIS RESULTS

Investment Horizon	5 years	10 years	15 years	20 years
Savings-Investment Ratio (SIR)	2.43	4.64	6.64	8.46
Net Present Value (NPV)	\$243,391	\$618,533	\$958,965	\$1,267,897

GREENHOUSE GAS EQUIVALENTS

	Value	Unit
Annual carbon dioxide avoided	138,950	pounds
Tree seedlings grown for 10 years	1616	tree seedlings
Equivalent annual home electricity use	9	homes

DISCLAIMER

Copyright by BELIMO Aircontrols (USA) Inc. Danbury, CT 2014. This is an estimating tool. Results are dependent on the accuracy of input values and specific plant.

The Home tab end to end. Everything else in the workbook is downstream of this page.

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Project identification

FIELD	CELL	WHAT TO ENTER
Project	F7	Free text. Job or building name — appears on printouts.
Information	F8	Free text. A one-line description, e.g. “4 chillers; installed 1600 tons”.
Climate region	F9	Dropdown. Picks the cooling-degree-day profile from the Energy tab.
Chiller cooling	J9	Dropdown — Water Cooled or Air Cooled.
Plant design	N9	Dropdown — Primary/Secondary or Variable Primary.
Utility Company	F10	Dropdown. Choosing a utility with a supported programme reveals its rebate tab.
Program	J10	Dropdown. Populated from the utility above. ConEd’s C&I programme adds the two ConEd tabs.

Input data: chilled water plant

FIELD	CELL	UNIT	WHAT IT IS
Electricity cost (average)	L12	\$/kWh	Blended all-in rate. Everything priced in dollars runs through this one number.
Design cooling load	L13	tons	Plant capacity at design conditions.
Number of chillers	L14	pcs.	How many machines share the load.
Operating hours per day	L15	hours	With days per week, sets annual run hours.
Operating days per week	L16	days	
Chiller design efficiency	L17	kW/ton	Full-load kW/ton from the chiller submittal — not IPLV. Becomes KW1 on the Chiller tab.
Actual Secondary Pump Head	L18	ft	Head the distribution pumps work against at design flow.
Pump, VFD and motor efficiency	L19	%	Combined wire-to-water efficiency.
Design Delta-T	L20	°F	What the plant was designed to achieve.
Actual Delta-T before Energy Valve	L21	°F	What it actually runs at today — the “before” case.
Future Energy Valve Delta-T setting	L22	°F	What you expect to hold with Energy Valves — the “after” case.

The three Delta-T fields are the heart of the estimate. Actual and Future are the two cases the whole comparison is built from, and the tool enforces a gap between them so the pair always makes physical sense.

Annual energy costs

Calculated — nothing to type. One row per Delta-T from your Actual up through your Future setting, colour-graded red (worst) to green (best): average flow, average tonnage, pumping cost, chiller cost, total cost and annual energy. The summary line beneath is the number most customers care about — the percentage and dollar reduction for moving from the Actual Delta-T to the Future one.

Input financials

FIELD	CELL	UNIT	WHAT IT IS
Incremental Energy Valve investment	L35	\$	Incremental is the operative word – the extra cost of Energy Valves over the valves that would have been fitted anyway, not the whole valve package.
Discount rate	L36	%	Feeds every present-value calculation. See section 7.3.
Annual energy-cost inflation	L37	%	Escalates the savings year on year, because energy prices rise.

Financial analysis results and greenhouse gas equivalents

The **Savings-Investment Ratio (SIR)** and **Net Present Value (NPV)** at 5, 10, 15 and 20 years, colour-graded, with the full workings on the Financials tab. Both are explained properly in section 7.3 - for now, treat SIR as how many dollars back per dollar in, and NPV as what the project is worth in today's money.

Below them the emissions avoided, expressed as pounds of CO₂ and as the two equivalents most people can picture - tree seedlings grown for ten years, and homes' annual electricity use.

4 Energy tab

Where the year gets divided into load bins, and where the climate profile lives.

BELIMO
Home
Energy
Chiller
Pump
Financials
Results

Operating Hours / Day	24	hours
Operating Days / Week	7	days
Annual Operating Hours	14955	hours

Editable field

Label

Point	Bin	Load [%]	Time [%]	Operating Hours
A	Design	100%	5.0%	748
B	Peak	75%	15.0%	2243
C	Low	50%	45.0%	6730
D	Min	25%	35.0%	5234
				14955

Operating hours, the four load bins, and the regions map. The two grey boxes are the on-sheet key for editable versus label cells.

Load bins

Rather than simulate 8,760 hours, the tool uses four representative operating points and assigns each a share of the year. Only the Time [%] column is editable.

POINT	BIN	LOAD	CELL	MEANING
A	Design	100%	E10	Full-load hours — the design day.
B	Peak	75%	E11	Hot-weather part load.
C	Low	50%	E12	The bulk of a typical year.
D	Min	25%	—	Remainder — calculated so the four shares total 100%.

Change these only with a reason. The defaults are a reasonable commercial-office profile. If you have real trend data or a load profile from the BAS, use it — otherwise leave them alone, because the bin split drives every hour, kWh and dollar downstream.

Climate regions

Climate region	Cooling Degree Days [°F·day]	Humidity Adjustment	Multiplier
North East	1034	30%	0.88
Central	1090	30%	0.93
East North Central	975	30%	0.83
West North Central	1123	15%	0.85
South East	2264	30%	1.93
South	2004	30%	1.71
South West	2750	5%	1.89
Mountain	1450	5%	1.00
North West	720	5%	0.50
West	1850	5%	1.27
Custom Region	3793	30%	3.23
Average US	1526	19%	1.19

Selected Value

1.71

Each region carries cooling degree days, a humidity adjustment and the resulting multiplier. The Selected Value at the bottom is the one your Home-tab region choice pulled through.

ROW	CELLS	USE
Custom Region	E31, F31	Editable. Enter your own cooling degree days and humidity adjustment for a site that does not fit a standard region – then select “Custom Region” on the Home tab.
Average US	—	Reference only – the national average, for sanity-checking.

5 Chiller tab

Breaks total plant efficiency into its six components, then works out chiller energy at each Delta-T.

Home
Energy
Chiller
Pump
Financials
Results

Aggregation of energy consumption

	[kW/ton]	Ratio
KW1 Compressor	0.630	83%
KW2 Constant Speed Primary Pump	0.038	5%
KW3 Condenser Pump	0.040	5%
KW4 Cooling Tower Fan	0.040	5%
KW5 Chiller Controls	0.005	1%
KW6 Transmission Oil Pump and Heater	0.010	1%
Total Chiller Plant Energy Consumption	0.763	100%

Chiller Savings Method:

Original (auxiliaries only)

Chiller Cycling Percent:

0%

Chiller energy usage at Delta-T [°F]:

6

Load [%]	Time [%]	Time [h]	Flow [GPM]	tons	Chillers On	Power [kW]
100	5.0	748	3000	750	4	199
75	15.0	2243	3000	750	4	199
50	45.0	6730	3000	750	4	199
25	35.0	5234	1500	375	2	100

Chiller costs

\$294,654

Avg Tonage

619

Plant kW/ton

0.265

Energy [kWh]

2455448

The kW/ton build-up, the savings method, the cycling allowance, and the first of several per-Delta-T calculation blocks.

Aggregation of energy consumption

Plant efficiency is not just the compressor. This table adds the auxiliaries to get a true plant kW/ton, and shows each as a share of the total.

COMPONENT	CELL	SOURCE
KW1 Compressor	—	Calculated — your Chiller design efficiency (Home L17). Change it there.
KW2 Constant Speed Primary Pump	—	Calculated — primary/secondary plants only.
KW3 Condenser Pump	F8	Editable. Default 0.040 kW/ton. Water-cooled plants only.
KW4 Cooling Tower Fan	F9	Editable. Default 0.040 kW/ton. Water-cooled plants only.
KW5 Chiller Controls	F10	Editable. Default 0.005 kW/ton.
KW6 Transmission Oil Pump and Heater	F11	Editable. Default 0.010 kW/ton.
Total Chiller Plant Energy Consumption	—	The sum — the plant kW/ton actually used in the costing.

What KW1–KW6 mean and where to get them

The total is multiplied by the tons of chiller capacity running in each load bin, so every figure has to be on the same basis: kilowatts drawn per ton of chiller capacity while that chiller runs. Divide equipment dedicated to one chiller by that chiller's rated tons, and plant-wide equipment by plant capacity. The kilowatts come from the motor nameplate ($\text{HP} \times 0.746 \times \text{load factor} \div \text{motor efficiency}$, with a real load

factor of 0.75–0.85), the pump schedule ($\text{GPM} \times \text{head in ft} \div (5,308 \times \text{overall efficiency})$), or a measurement ($\text{volts} \times \text{amps} \times 1.732 \times \text{power factor} \div 1,000$, or a power meter or VFD display).

KW1 Compressor. Electricity the chiller draws per ton at full load — the largest term by far. Take it from the chiller submittal or AHRI certificate (full-load kW/ton), or input kW $\div$ rated tons from the nameplate; on site, a chiller kW trend $\div$ tons, where tons = $\text{GPM} \times \Delta T \div 24$. Never IPLV: it is a part-load average and understates energy.

KW2 Constant-speed primary pump. The pumps pushing water through each chiller on a primary/secondary plant. It cannot be entered: the tool assumes they use the same energy as the secondary pumps at full speed, from design flow and the pump head and efficiency on Home, $\div$ design tons — 3,000 GPM at 80 ft and 80% gives 56.5 kW, $\div$ 1,500 tons = 0.038. Primary loops often run at lower head, so it can be on the high side; to sanity-check, divide a primary pump's motor kW by the rated tons of the chiller it serves.

KW3 Condenser pump. Moves condenser water between the chiller and the cooling tower; water-cooled plants only. From the condenser pump schedule (GPM and head) or the motor nameplate, $\div$ the tons of the chiller it serves. About 3 GPM per ton at 60–80 ft gives 0.04–0.055 — for example $1,125 \text{ GPM} \times 70 \text{ ft} \div (5,308 \times 0.75) = 19.8 \text{ kW}$, $\div$ 375 tons = 0.053.

KW4 Cooling tower fan. The fan motors that reject the heat; water-cooled plants only. From the tower submittal or nameplates — total every fan, convert to kW, $\div$ the tons the tower serves. Typically 0.02–0.06, with fixed- or two-speed fans at the high end. The tool scales this with chillers running, not with fan speed, so it cannot credit a tower VFD.

KW5 Chiller controls. The control panel, sensors, actuators and starter electronics. From the control-panel electrical data on the submittal — usually 0.5–2 kW per chiller — $\div$ rated tons. Typically 0.002–0.005; it rarely matters.

KW6 Transmission oil pump and heater. The oil pump that lubricates the compressor bearings and, on gear-driven centrifugals, the gearbox — plus the oil sump heater. From the auxiliary electrical data on the submittal: oil pump motor (often 1–3 HP) plus heater (often 1–3 kW), $\div$ rated tons. Oil-free chillers with magnetic bearings have neither: enter 0.

Double counting. A manufacturer's full-load rating usually covers the whole chiller package, often including its controls and oil pump. Check the submittal footnotes — if they are included, KW5 and KW6 overlap KW1 slightly, about 2%.

Easy to inflate. With Original, the default, KW2–KW6 are the whole chiller-side saving. On the shipped example every +0.01 kW/ton on KW3, KW5 or KW6 adds about 7% to it, and on KW4 nearly 10% (under Full plant, about 1.3%). Leave the defaults unless you have submittal data or measurements.

Chiller Savings Method

Editable dropdown (F13, default Original). Decides how much of the chiller plant's energy the better Delta-T is credited with. The pump saving is the same either way.

OPTION	WHAT IS CREDITED	USE IT WHEN
Original (auxiliaries only)	The method of the original 2021 estimator. The building still needs the same tons, so compressor energy is not counted — only the primary and condenser pumps, tower fans, controls and oil pumps that stop with fewer chillers running.	The default. The figure to put in front of a customer or a utility — the next section explains why.
Full plant (incl. compressor)	Every chiller the plant no longer needs to run is credited its full energy — compressor included, at the full-load kW/ton from Home — plus its pumps, tower fans, controls and oil pump.	Only as an upper bound, and say so. Most of its extra saving is compressor energy that is not really saved.

It moves the answer a lot. On the shipped example the total saving is \$89,357 a year with Original and \$466,038 with Full plant — the chiller side is \$60,342 against \$437,023, while the pump side stays at \$29,014. Say which method a figure uses whenever you quote it.

Original is not a straight copy of the 2021 formula. Two errors are corrected: tower-fan savings count only for water-cooled plants, and KW5 and KW6 are multiplied by tons rather than by the number of chillers. On the same plant the tool lands close to the 2021 figure, not exactly on it. Like every field, the dropdown has a blue i button that explains both options in the selected language.

Compressor versus auxiliaries — why Original is the default

Two kinds of equipment use electricity in a chiller plant, and a better Delta-T affects them very differently.

The compressor is the refrigeration machine inside each chiller — where the cooling is actually made. Its power follows the tons of cooling the building needs at that moment. It is KW1: 0.630 kW/ton on the example, 83% of plant power.

The auxiliaries are the supporting equipment that runs because a chiller is switched on, however hard that chiller is working — KW2–KW6.

TERM	EQUIPMENT	KW/TON	SHARE
KW2	Constant-speed primary pump (one per chiller)	0.038	5%
KW3	Condenser pump	0.040	5%
KW4	Cooling tower fans	0.040	5%
KW5	Chiller controls	0.005	1%
KW6	Oil pump and oil heater	0.010	1%
Total	Auxiliaries	0.133	17%

What happens when Delta-T improves. Take the 100% load bin of the example: the building needs 750 tons, and each chiller is rated 375 tons. At today's low Delta-T the plant has to push 3,000 GPM, so all four chillers run. At the Energy Valve Delta-T the same 750 tons needs half the flow, so two chillers run.

POWER DRAWN	4 CHILLERS ON	2 CHILLERS ON	SAVING
Auxiliaries (0.133 kW/ton × 375 t per chiller)	200 kW	100 kW	100 kW — real. The pumps, fans and heaters of the idle chillers stop.
Compressors, as Full plant counts them	945 kW	472 kW	472 kW
Compressors, physically	about 472 kW	about 472 kW	About zero — the same 750 tons is made either way.

Full plant charges every running chiller its full-load compressor power ($4 \times 375 \times 0.63 = 945$ kW). But the building still needs only 750 tons, so the compressors make 750 tons whichever machines run: about $750 \times 0.63 = 472$ kW. Switching two chillers off moves that load onto the other two; it does not make it go away.

Why Original is the default. Most of the Full plant saving is compressor energy that is not really saved. Original gives the figure an engineer can defend, and it is the one to put in front of a customer or a utility. Some compressor effect is real — a chiller idling at very low load often runs at a worse kW/ton — but it is a few percent of compressor energy, it can go either way on variable-speed chillers, and it is nowhere near half.

The tables above illustrate the principle rather than reproduce the tool's exact arithmetic — the tool, for instance, scales cooling-tower fans with water flow rather than with chillers running.

Chiller Cycling Percent

Editable (F15, default 0%). An allowance for the efficiency penalty of chillers short-cycling at low load. Raise it if the plant is known to cycle badly.

Per-Delta-T blocks

Below that, one block per Delta-T in the comparison range. Each is headed by its Delta-T and shows the four bins with load, time, flow, tons, how many chillers are running and the resulting power — then chiller cost, average tonnage, plant kW/ton and annual kWh. All calculated.

The same idea for the distribution pumps – flow, head and power in each bin, at each Delta-T.

Home

Energy

Chiller

Pump

Financials

Pump design factor

100%

Design chilled-water distribution flow

12

Pump Affinity Laws

$$\frac{RPM_1}{RPM_2} = \frac{GPM_1}{GPM_2} \quad \frac{HEAD_1}{HEAD_2} = \left(\frac{RPM_1}{RPM_2}\right)^2 \quad \frac{HP_1}{HP_2} = \left(\frac{RPM_1}{RPM_2}\right)^3$$

$$HP = \frac{GPM \times HD \times \text{Specific Gravity}}{3960 \times \text{Pump Efficiency}}$$

Editable field

Label

Load [%]	Time [%]	Time [h]	Flow [GPM]	Head [ft]	Power [HP]	Power [kW]
100	5.0	748	3000	80	75.8	
75	15.0	2243	2250	45	32.0	
50	45.0	6730	1500	20	9.5	
25	35.0	5234	1061	10	3.3	

Pump costs

Annual Gallons

Avg GPM

Energy [kWh]

\$18,763

1376230268

1534

156360

57

Minimum pump head

[ft]

10

Load [%]	Time [%]	Time [h]	Flow [GPM]	Head [ft]	Power [HP]	Power [kW]
			1061	10	3.3	2.5

Chilled-water distribution flow at Delta-T [°F]:

6

Load [%]	Time [%]	Time [h]	Flow [GPM]	Head [ft]	Power [HP]
100	5.0	748	3000	80	75.8
75	15.0	2243	3000	80	75.8
50	45.0	6730	3000	80	75.8
25	35.0	5234	1500	20	9.5

Pump costs

Annual Gallons

Avg GPM

Energy [kWh]

\$70,362

2220839104

2475

586352

Design case at the top, then a block per Delta-T. Pump costs, annual gallons, average GPM and energy sit to the right of each block.

FIELD	CELL	WHAT IT IS
Pump design factor	F5	Editable, default 100%. A safety/oversizing multiplier on design flow. Leave at 100% unless you know the pumps were deliberately oversized.
Minimum pump head	F13	Editable, default 10 ft. A floor on head as flow falls. Real systems never reach zero head — static lift, control valves and minimum-speed limits keep it above zero, and without a floor the affinity laws would overstate savings at low load.
Bin tables	—	Calculated. Load, time, flow, head and horsepower per bin, plus cost, annual gallons, average GPM and kWh.

Why pumping savings look large. Pump power varies with roughly the cube of flow. Cutting flow by a third cuts pump power by about 70%. That is why the pumping percentage saving on the Home tab is usually much bigger than the chiller percentage — though in absolute dollars the chiller side is normally the larger prize.

7 Financials tab

The full ten-year cash-flow model behind the headline numbers on Home. This is the tab that gets misread most often, so it is worth understanding properly.

Simulating Streams of Cash Inflows and Outflows											
Discount rate:	4%										
Finance rate:	6%										
Reinvestment rate:	6%										
Inflation Rate:	2%										
Today	End of Year 1	End of Year 2	End of Year 3	End of Year 4	End of Year 5	End of Year 6	End of Year 7	End of Year 8	End of Year 9	End of Year 10	
Date:	0	1	2	3	4	5	6	7	8	9	10
CASH OUTFLOWS											
Single Investment	(\$170,000)	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Phased Investment	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Financed Investment	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Subtotal	(\$170,000)	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Rebate Rec'd at Project Start	\$0										
Subtotal Outflows	(\$170,000)	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
CASH INFLOWS											
Energy savings	\$89,357	\$91,144	\$92,967	\$94,826	\$96,723	\$98,657	\$100,630	\$102,643	\$104,696	\$106,790	
Maintenance/Penalty Savings	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	
Rebates Rec'd after Project Start	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	
Subtotal Inflows	\$89,357	\$91,144	\$92,967	\$94,826	\$96,723	\$98,657	\$100,630	\$102,643	\$104,696	\$106,790	
Annual cash flow	(\$170,000)	\$89,357	\$91,144	\$92,967	\$94,826	\$96,723	\$98,657	\$100,630	\$102,643	\$104,696	\$106,790
PV of Outflows (for SIR)	(\$170,000)	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
PV of Inflows (for SIR)	\$0	\$85,920	\$84,268	\$82,647	\$81,058	\$79,499	\$77,970	\$76,471	\$75,000	\$73,558	\$72,143
Annual Present Value	(\$170,000)	\$85,920	\$84,268	\$82,647	\$81,058	\$79,499	\$77,970	\$76,471	\$75,000	\$73,558	\$72,143
Note: Certain Returns Vary Depending on the Length of the Analysis Term											
	10 Year	1 Year	2 Years	3 Years	4 Years	5 Years	6 Years	7 Years	8 Years	9 Years	10 Years
Net Present Value (NPV)	\$618,533	(\$84,080)	\$188	\$82,835	\$163,892	\$243,391	\$321,361	\$397,832	\$472,832	\$546,390	\$618,533
Simple Payback Period (SPP)	1.9										
Return on Investment (ROI)	52.6%										
Internal Rate of Return (IRR)	53.7%	-47.4%	4.1%	27.9%	39.6%	45.7%	49.2%	51.2%	52.5%	53.2%	53.7%
Mod. Internal Rate of Return (MIRR)	22.3%	-47.4%	4.6%	19.5%	24.0%	25.2%	25.1%	24.6%	23.9%	23.1%	22.3%
Savings-Investment Ratio (SIR)	4.6	0.5	1.0	1.5	2.0	2.4	2.9	3.3	3.8	4.2	4.6

The whole tab: rates top-left, cash flows across ten years, then the metric block at the bottom.

7.1 The four rates

RATE	CELL	SET WHERE	WHAT IT DOES
Discount rate	B9	Home L36	Converts future dollars into today's dollars. Usually the organisation's cost of capital or required return. Raising it lowers both headline results, because distant savings count for less.
Finance rate	B10	Editable here	What it costs to borrow the money. Used only by the Modified Internal Rate of Return (section 7.3).
Reinvestment rate	B11	Editable here	What the savings earn once received. Used only by the Modified Internal Rate of Return.
Inflation rate	B12	Home L37	Escalates the annual energy saving. This is why the year-10 saving is larger than the year-1 saving.

7.2 The cash-flow grid

	Today	End of Year 1	End of Year 2	End of Year 3	End of Year 4	End of Year 5	End of Year 6	End of Year 7	End of Year 8	End of Year 9	End of Year 10
Date:	0	1	2	3	4	5	6	7	8	9	10
CASH OUTFLOWS											
Single Investment	(\$170,000)	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Phased Investment	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Financed Investment	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Subtotal	(\$170,000)	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Rebate Rec'd at Project Start	\$0										
Subtotal Outflows	(\$170,000)	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
CASH INFLOWS											
Energy savings	\$89,357	\$91,144	\$92,967	\$94,826	\$96,723	\$98,657	\$100,630	\$102,643	\$104,696	\$106,790	
Maintenance/Penalty Savings	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Rebates Rec'd after Project Start	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Subtotal Inflows	\$89,357	\$91,144	\$92,967	\$94,826	\$96,723	\$98,657	\$100,630	\$102,643	\$104,696	\$106,790	
Annual cash flow	(\$170,000)	\$89,357	\$91,144	\$92,967	\$94,826	\$96,723	\$98,657	\$100,630	\$102,643	\$104,696	\$106,790
PV of Outflows (for SIR)	(\$170,000)	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
PV of Inflows (for SIR)	\$0	\$85,920	\$84,268	\$82,647	\$81,058	\$79,499	\$77,970	\$76,471	\$75,000	\$73,558	\$72,143
Annual Present Value	(\$170,000)	\$85,920	\$84,268	\$82,647	\$81,058	\$79,499	\$77,970	\$76,471	\$75,000	\$73,558	\$72,143

Columns run Today, then End of Year 1 through End of Year 10. Money out is negative; money in is positive.

ROW	EDITABLE?	MEANING
Single Investment	Years 1–10	The up-front cost. Today's column comes straight from Home L35; later columns let you add a further lump.
Phased Investment	All columns	For work spread over several years – a floor a year, say.
Financed Investment	All columns	For a financed package, entered as the payments fall.
Rebate Rec'd at Project Start	Today only	Incentive money in hand at day zero. Reduces the net outflow.
Subtotal Outflows	Calculated	All investment, less any start rebate.
Energy savings	Calculated	From the plant model, escalated each year by the inflation rate.
Maintenance/Penalty Savings	Year 1	Anything else the valves save – avoided maintenance, avoided utility penalties.
Rebates Rec'd after Project Start	Years 1–10	Incentives paid later, e.g. on measured performance.
Subtotal Inflows	Calculated	Everything coming back.
Annual cash flow	Calculated	Inflows plus outflows for that year. The series every metric is built from.
PV of Outflows (for SIR) / PV of Inflows (for SIR)	Calculated	Each side discounted to today. Used for the Savings-Investment Ratio. The Outflows row carries an i button explaining present value.
Cumulative Cash Flows	Calculated	Running total, undiscounted. The year it turns positive is the true payback year.
Annual Present Value	Calculated	That year's cash flow in today's money.
Cumulative Present Value	Calculated	Running total of the above - the last figure is the Net Present Value.

7.3 What the terms mean

	10 Year	1 Year	2 Years	3 Years	4 Years	5 Years	6 Years	7 Years	8 Years	9 Years	10 Years
Net Present Value (NPV)	\$618,533	(\$84,080)	\$188	\$82,835	\$163,892	\$243,391	\$321,361	\$397,832	\$472,832	\$546,390	\$618,533
Simple Payback Period (SPP)	1.9										
Return on Investment (ROI)	52.6%										
Internal Rate of Return (IRR)	53.7%	-47.4%	4.1%	27.9%	39.6%	45.7%	49.2%	51.2%	52.5%	53.2%	53.7%
Mod. Internal Rate of Return (MIRR)	22.3%	-47.4%	4.6%	19.5%	24.0%	25.2%	25.1%	24.6%	23.9%	23.1%	22.3%
Savings-Investment Ratio (SIR)	4.6	0.5	1.0	1.5	2.0	2.4	2.9	3.3	3.8	4.2	4.6

V.2026.11.0

The six metrics, each shown at every horizon from one to ten years.

PV

Present Value

What a future amount of money is worth **today**. A dollar saved in year eight is not worth a dollar now — you could have invested that dollar in the meantime. Present value shrinks each future amount back accordingly.

How the tool gets it: $\text{cash flow} \div (1 + \text{discount rate})^{\text{years}}$. Everything else on this tab is built from it.

NPV

Net Present Value

Add up the present value of every year — including the money spent on day one, which is negative — and you have the NPV. It is the single best answer to “is this project worth doing?”

NPV > 0 → worth doing. NPV < 0 → not at this discount rate.

Positive means the project returns more than the discount rate you demanded, and the NPV is the surplus, in today’s dollars. Negative does not mean it loses money — it means it does not clear the bar you set.

How the tool gets it: the sum of the Annual Present Value row across Today plus all ten years.

SIR

Savings-Investment Ratio

The present value of everything coming in, divided by the present value of everything going out. A ratio rather than an amount, so it compares projects of different sizes on equal terms.

SIR > 1.0 → returns more than it costs. SIR < 1.0 → does not pay back in present-value terms.

An SIR of 1.8 means every dollar invested returns \$1.80 in today’s money. Many public-sector and utility incentive programmes require a minimum SIR, which is why it appears on the Home tab alongside NPV.

How the tool gets it: $\text{PV of Inflows} \div |\text{PV of Outflows}|$, both summed across the horizon.

SPP

Simple Payback Period

How many years of savings it takes to repay the investment, ignoring the time value of money entirely. The number everyone asks for first.

How the tool gets it: $\text{investment} \div \text{first-year cash flow}$. Note it uses the first year only — because savings escalate with inflation, the real payback usually arrives slightly sooner than SPP suggests. Use the Cumulative Cash Flows row to see the actual crossover year.

ROI

Return on Investment

Here, the **first-year** return: what the project returns in year one as a percentage of what it cost.

How the tool gets it: $\text{year-1 cash flow} \div \text{investment}$. This is NOT a lifetime return, which is what people usually assume the term means. In the worked example it reads 274% - meaning year one alone returns about 2.7 times the outlay, not that the project earns 274% a year for its whole life. Quote it carefully, or quote the Internal Rate of Return instead.

IRR

Internal Rate of Return

The discount rate at which the NPV would come out exactly zero — in effect the interest rate the project itself earns. Compare it against your hurdle rate: if the IRR beats the rate you could get elsewhere, the project is attractive.

How the tool gets it: Excel's IRR over the annual cash flow row. Its weakness: IRR quietly assumes every dollar of savings is reinvested at the IRR itself, which is rarely true — which is exactly what MIRR fixes.

MIRR

Modified Internal Rate of Return

On the sheet this row reads “Mod. Internal Rate of Return (MIRR)” — abbreviated only so the label and its i button both fit the column.

IRR's more honest sibling. It separates the two rates IRR conflates: what borrowing costs you, and what the returning savings actually earn. Usually the more defensible figure to put in front of a finance team.

How the tool gets it: Excel's MIRR over the annual cash flow row, using the Finance rate (B10) and Reinvestment rate (B11) you set at the top of this tab. Set those two honestly and MIRR becomes the number worth arguing from.

Which one should you lead with? For a building owner, simple payback and NPV. For a finance department, NPV and MIRR. For a utility or public-sector programme, SIR — it is often written into the eligibility rules.

7.4 Reading the horizon columns

Every metric is repeated across columns headed 1 Year through 10 Years, with a summary figure at the left. Each column answers the same question over a different holding period: if we judged this project over N years, how would it look?

Short horizons look poor and long horizons look better, because the investment lands entirely in year zero while the savings accumulate. That is expected. Quote the horizon that matches how the customer actually evaluates capital — and say which horizon you are quoting.

The worked example, read across. A \$170,000 investment against \$89,357 of annual saving pays back in 1.9 years. The one-year column shows a Net Present Value of -\$84,080, two years roughly breaks even at \$188, and the figure rises to \$618,533 at ten. That crossover from negative to positive is exactly what the columns are there to reveal — so quote a horizon the customer actually uses.

7.5 Built-in help - the blue i buttons

Every term on this tab now explains itself. You do not need this manual open to answer “what does that number mean?” in front of a customer.

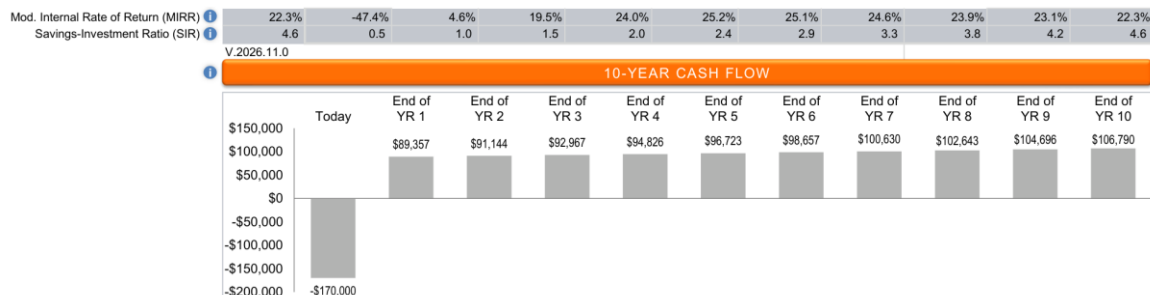
Each metric row carries a small blue i button just after its label. Click it and a plain-language explanation opens in whichever language the tool is set to. The explanations are written for someone who does not work in finance, and each one closes with a worked example built from your figures rather than a generic illustration - so what the customer reads always matches what is on screen.

	10 Year	1 Year	2 Years	3 Years	4 Years	5 Years	6 Years	7 Years	8 Years	9 Years	10 Years
Net Present Value (NPV) ⓘ	\$618,533	(\$84,080)	\$188	\$82,835	\$163,892	\$243,391	\$321,361	\$397,832	\$472,832	\$546,390	\$618,533
Simple Payback Period (SPP) ⓘ	1.9										
Return on Investment (ROI) ⓘ	52.6%										
Internal Rate of Return (IRR) ⓘ	53.7%	-47.4%	4.1%	27.9%	39.6%	45.7%	49.2%	51.2%	52.5%	53.2%	53.7%
Mod. Internal Rate of Return (MIRR) ⓘ	22.3%	-47.4%	4.6%	19.5%	24.0%	25.2%	25.1%	24.6%	23.9%	23.1%	22.3%
Savings-Investment Ratio (SIR) ⓘ	4.6	0.5	1.0	1.5	2.0	2.4	2.9	3.3	3.8	4.2	4.6
V.2026.11.0											

One button per metric, lined up down the label column.

BUTTON	WHAT IT EXPLAINS
PV of Outflows (for SIR)	Present value - why a dollar received in five years is worth less than a dollar today, and how the discount rate converts one into the other.
Net Present Value (NPV)	What the project is worth today after subtracting the investment, and what the horizon columns are doing.
Simple Payback Period (SPP)	How long the savings take to repay the spend - and, importantly, what SPP ignores.
Return on Investment (ROI)	Total gain as a percentage of the spend, and why timing is invisible to it.
Internal Rate of Return (IRR)	The effective annual return, how to compare it with your cost of money, and why it flatters a project.
Mod. Internal Rate of Return (MIRR)	Why MIRR is the more defensible figure, and which two rates it uses.
Savings-Investment Ratio (SIR)	Dollars saved per dollar invested, with the 1.0 break-even rule.

The four chart bands lower down the tab each have their own i button too, sitting to the left of the orange title bar.



The button sits just left of the section title.

SECTION	WHAT THE EXPLANATION COVERS
10-Year Cash Flow	Real money in and out each year, before any discounting. The first bar is the investment.
Cumulative 10-Year Cash Flow	The running total. Where it crosses zero is the payback point.
10-Year Present Value	The same savings converted into today's money - which is why later bars shrink even as the savings grow.
Cumulative 10-Year Present Value	The running total in today's money. The final bar is the 10-year NPV - the same number as the NPV row above.

They never print. The i buttons are screen-only. They do not appear on a printed page or in an exported PDF, so the report a customer receives stays clean.

They are not only on this tab. The same buttons appear beside the input fields on Home, Energy, Chiller and Pump - including one beside the Pump Affinity Laws diagram, which explains the relationship between flow, head and pump power. If a field has a button, it has an explanation.

8 Results tab

The customer-facing summary. Nothing to fill in — this is the page you present or print.

Results

How Energy Valves Deliver Savings

Explore a different Delta-T: From: 8.0 °F To: 12.0 °F

By raising the chilled water Delta-T from 8.0 to 14.0 °F, Energy Valves reduce the flow rate needed to meet the same cooling load, cutting pump energy, while letting the chiller run more efficiently, reducing chiller energy too. Together, these two effects drive the total plant savings shown below.

Total Annual Savings

\$77,030

Total Annual Energy Saved

641,913 kWh

Metric	dT = 8.0 °F	dT = 12.0 °F	Estimated annual savings	
Delta-T	8	12	Pumping savings (from reduced pump flow)	\$22,648
Average flow	2,006	1,534	Chiller savings (from improved chiller efficiency)	\$54,382
Average cooling load	668.8	766.9	Total cost savings	\$77,030
Pumping cost per year	\$41,411	\$18,763	Energy savings	641,913
Chiller cost per year	\$249,763	\$195,381		
Total cost per year	\$291,174	\$214,144		
Energy per year	2,426,450	1,784,536		

dT = 8.0 °F — Layman's Interpretation

Current pumping summary: \$41,411/yr; flow 2,006.3 GPM

Current annual chiller cost: \$249,763

dT = 12.0 °F — Layman's Interpretation

Proposed pumping summary: \$18,763/yr; flow 1,533.7 GPM

Proposed annual chiller cost: \$195,381

Headline savings, the before/after comparison, and the plain-language interpretation.

The Delta-T explorer

At the top sits a spinner beside “Explore a different Delta-T”. It opens on your design Delta-T. Click the arrows and the whole comparison — flow, tonnage, pumping cost, chiller cost, total cost and energy — re-runs against that scenario while the customer watches. The “From” case stays fixed at the plant’s actual Delta-T.

It is the most persuasive thing on the tab: it turns a static estimate into a live answer to “what if we only got halfway there?”

What each block shows

BLOCK	CONTENTS
Headline savings	Total annual dollars saved and total annual kWh saved, in large type.
Comparison table	Actual versus Custom side by side: Delta-T, average flow, average cooling load, pumping cost, chiller cost, total cost, energy — with the savings column breaking the difference into pumping, chiller, total and energy.
Layman's interpretation	The same result written as sentences, in the selected language — for pasting into a proposal.
Pump bin details	The four bins side by side for both cases, with flow, head and pump power.

Pump Bin Details – dT = 8.0 °F				
Load [%]	Time [h]	Flow GPM	Head ft	Pump HP
100	748	3,000	80.0	75.8
75	2,243	3,000	80.0	75.8
50	6,730	2,250	45.0	32.0
25	5,234	1,125	11.3	4.0

Pump Bin Details – dT = 12.0 °F				
Load [%]	Time [h]	Flow GPM	Head ft	Pump HP
100	748	2,571	58.8	47.7
75	2,243	1,929	33.1	20.1
50	6,730	1,286	14.7	6.0
25	5,234	1,061	10.0	3.3

Financial & Environmental Highlights	
Net Present Value (NPV)	\$509,752
Savings-Investment Ratio (SIR)	4.00
Simple Payback Period (SPP)	2.2 years
Annual carbon dioxide avoided	119,781 pounds
Tree seedlings grown for 10 years	1,393 tree seedlings
Equivalent annual home electricity use	7 homes

Utility Rebate Support	
Energy savings	641,913 kWh
Annual carbon dioxide avoided	119,781 pounds

This analysis uses a bin-based energy model built on region-specific weather data and standard pump affinity-law calculations, providing a defensible basis for utility rebate and incentive program verification.

Financial highlights, greenhouse-gas equivalents, and the rebate-support block with its methodology note.

The Utility Rebate Support block restates energy saved and CO₂ avoided in the form incentive programmes ask for, and closes with a methodology note describing the bin-based model and weather data – there so a programme reviewer can see the basis without having to ask.

9 ConEd Rebate tabs

Two extra tabs that appear only when you select Con Edison and its C&I Energy Efficiency programme on the Home tab.

Con Edison C&I Rebate Estimate

The Energy Valve qualifies under Building Automation Systems (program manual §2.2.6). Con Edison applies caps this tool does not, so the figures here are what they will actually credit — not what the Home sheet shows. English and US units throughout, because the C&I program is New York City only.

Project classification

Building type	Hospital
HVAC system	VAV econ
Building vintage	Prior to 1979

Chilled water pump motor

Pump motor nameplate HP	75.0
Number of motor poles	4 Pole - 1,800 RPM
Enclosed or open motor	Enclosed

Con Edison lookups

TRM classification	Large Commercial	
Distribution type used	VAV econ	
Equivalent full load hours (EFLH)	1,129	hours
Operating hours Con Edison will use	1,129	hours
Pump motor efficiency	95.4%	

Con Edison savings calculation

Design pump flow	3,000	GPM
Average existing flow	2,006	GPM
Average proposed flow	1,368	GPM
Flow reduction (capped at 30%)	30.0%	
Existing pump energy	16,758	kWh / yr
Pump energy savings	5,027	kWh / yr
Chiller baseline energy	1,066,905	kWh / yr
Chiller savings (0.5% cap)	5,335	kWh / yr
Total claimable savings	10,362	kWh / yr

Incentive

Incentive rate	\$0.35 / kWh	
Gross incentive	\$3,627	
Project cost	\$170,000	
Cap: 50% of project cost	\$85,000	
Estimated incentive	\$3,627	

Fill these in from the site survey.

The single biggest driver of the whole result — it sets the EFLH cap below. Must match Con Edison's list exactly.

Only used when the TRM class comes back as Large Commercial. Everything else falls back to n/a.

Only read when Building type is Multi-Family High-Rise. Ignored otherwise.

From the motor nameplate, not the calculated duty.

Nameplate horsepower, not brake HP. Round up to the next standard motor size.

With HP and enclosure this selects the efficiency Con Edison looks up. 4-pole covers most CHW pumps.

TEFC reads as Enclosed, ODP as Open.

Derived from their tables — nothing to enter here.

Con Edison's own mapping. Large Commercial gets an HVAC-specific EFLH; everything else uses n/a.

Dormitory is forced to Fan Coil; non-Large-Commercial is forced to n/a; otherwise your HVAC choice.

Cooling EFLH from Con Edison's table. Spans 0 h (parking garage) to 2,884 h (hotel).

MIN(EFLH, your actual hours). Your real run hours only count if they are LOWER than the table.

From Con Edison's motor table by HP + poles + enclosure. Falls back to 90% if the combination is not listed.

Their method, using your Home sheet inputs.

Design cooling load × 24 + design Delta-T, from the Home sheet.

Average flow at your Actual Delta-T, read from the Home sheet's Delta-T table.

Average flow at your Future Delta-T.

MIN(flow-based, Delta-T-based, 30%). Con Edison will not credit beyond 30% however good the Delta-T improvement.

Their model: $HP \times MIN(1, 0.75 \times (existing + design\ flow)^{2.7}) \times 0.746 \times hours + efficiency$. Note the 2.7 exponent, not the cube law.

Existing pump energy × flow reduction.

Design tons × kW/ton × hours. Matches Con Edison's Chiller Peak Load field.

Baseline × MIN(0.0012 per °F, 0.5%). Con Edison credits almost nothing on the chiller side — see the note below.

Pump + chiller. This is the number the rebate is paid on.

Building Automation Systems rate, program manual §2.2.6.

BAS electric rate. Steam is \$80/Mlbs, gas \$5/therm, fuel oil \$10/gal if those apply.

Claimable kWh × rate, before the cap.

Investment from the Home sheet. Con Edison's form wants labour and materials entered separately.

Incentives cannot exceed 50% of project cost, 100% of each measure cost, or \$1,000,000 per account per year.

The lower of gross incentive and the cap. This is what to quote.

The ConEd Rebate worksheet. The explanatory paragraph at the top states plainly that Con Edison applies caps this tool does not.

Read the figure on this tab as what ConEd will actually credit — not as a second opinion on the Home tab. Con Edison applies programme caps that the plant model does not, so this number is deliberately more conservative. The two are answering different questions. These tabs are English and US units only, by design: the C&I programme is New York City only.

Con Edison C&I Rebate Estimate

This estimate follows the calculation method of the Con Edison Commercial & Industrial Energy Efficiency Program for Building Automation Systems measures. Final incentive amounts are subject to Con Edison's review of the completed program application.

Project Information

Building Type	Hospital
HVAC System	VAV econ
Building Vintage	

Equipment

Pump Motor (HP)	75.0
Motor Poles	4 Pole - 1,800 RPM
Motor Enclosure	Enclosed

Estimated Program Savings

Equivalent Full Load Hours	1,129 hours
Pump Energy Savings	5,027 kWh / year
Chiller Energy Savings	5,335 kWh / year
Total Estimated Annual Savings	10,362 kWh / year

Estimated Incentive

Incentive Rate	\$0.35 / kWh
Project Cost	\$170,000
Program Cap (50% of Project Cost)	\$85,000

Estimated Annual Incentive

\$3,627

The ConEd Summary tab — a one-page version for the submission pack.

10 Printing and practicalities

PDF export

Each of the main tabs carries its own PDF button. It exports that tab to a PDF named after the sheet and the date and time, and asks you where to save it. The Results tab is normally the one you send.

Charts

The Home tab has three chart groups — annual energy costs, financial results, and a pair of pie charts showing where the plant's energy goes before and after. Each has a small button beside it to show or hide it, so you can keep the page compact until you want to talk to a chart.

If something looks wrong

SYMPTOM	CHECK
A cell will not let you type	It is a calculated result, not an input. Compare it against the shading key in section 2.
Savings look implausibly large	Check the gap between Actual and Future Delta-T, and the minimum pump head on the Pump tab. A very low head floor overstates low-load savings.
Every financial result looks wrong	Almost always the investment figure. Confirm Home L35 holds the incremental cost of the Energy Valves over the valves that would have been fitted anyway - not the whole valve package.
The ConEd tabs are missing	They only appear when Con Edison and the C&I programme are both selected on Home.
Numbers did not change after an edit	Press F9 to recalculate, or click away from the cell to commit the entry.

This tool produces an estimate. Results depend on the accuracy of the inputs and on plant-specific conditions — the disclaimer at the foot of the Home tab is the authoritative wording. Questions, corrections and feature requests go to Rick.Smith@us.belimo.com.

11 Release history

What changed in each release of the tool.

RELEASE	DATE	CHANGES
2026.11.1	2026-09-24	Added the standard admin keys used in every Belimo tool: F12 asks for the admin password and then shows and unprotects every sheet; F11 puts every sheet back exactly as it was. The Savings Estimator's sheets now use the same password as the other tools. Fixed: choosing the climate region South West or West turned every result into #N/A; all twelve regions now calculate. Corrected spelling mistakes in a chiller label and the help text.
2026.11.0	2026-09-22	Added automatic update checking through Belimo's update source. Added Check for Updates and Download Fresh Copy buttons to the About screen. Updates save through a normal Windows Save As dialog and never touch your current copy. The About screen also has a Download User Manual button, which fetches this manual.